

# Exam 3

個人成績報告

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## 題目詳情

### Q1 簡答題 （2分）

Give an example of how a 4-block file might be striped across 2 OSSs in Lustre.

你的答案

Block0：分配給OSS A

Block1：分配給OSS B

Block2：分配給OSS A循環輪詢

Block3：分配給OSS B循環輪詢

## Q2 簡答題 (2分)

What is Read-Only Replication (ROR) and how does it solve the consistency issue?

你的答案

只讀複製(read only replication)，僅複製不可變的文件，以避免一致性問題。此方法最簡單，也最容易保持一致，但只適合用於不需修改的檔案。

## Q3 簡答題 (2分)

How does the Primary Copy Protocol handle read and write operations?

你的答案

寫入集中由Primary處理，再廣播給Backup保證一致性。  
讀取可在任副本(Primary或Backup)進行升讀取效率。

## Q4 簡答題 (4分)

List four examples of file metadata.

你的答案

檔案大小，檔案權限與擁有者，時間戳記，檔案類型。

## Q5 簡答題 (4分)

In a central server algorithm for mutual exclusion, describe a situation where two requests might not be processed in the happened-before order.

你的答案

同一Process內部順序:若事件A在同一Process中早於事件B，則A->B  
訊息傳送順序，若事件A是送出message，事件B是接收該Message，則A->B  
遞移性，若A->B且B->C，則A->C

## Q6 簡答題 (3分)

Differentiate between External and Internal physical clock synchronization. Are externally synchronized clocks also internally synchronized?

你的答案

否，內部同步的時鐘並不一定需要外部同步。

### Q7 簡答題 (3分)

Explain the Quorum-Based Protocol for multi-copy updates. What conditions must be met for read and write quorums?

你的答案

$N_r$ 為讀取法定人數， $N_w$ 為寫入法定人數。

條件一： $N_r + N_w > N$

條件二： $N_w > N/2$

### Q8 簡答題 (3分)

Explain how a deadlock can occur in Maekawa's mutual exclusion algorithm using an example of three processes.

你的答案

有三個Process：P1, P2, P3其中 $V1=\{P1,P2\}$ ,  $V2=\{P2,P3\}$ ,  $V3=\{P3, P1\}$ 如果它們同時請求進入CS，則可能會發生P1投給自己不投給P2, P2投給自己不投給P3, P3投給自己不投給P1,結果每個Process都只能拿到一票，卻都等不到剩下來的票，形成了僵局。

### Q9 簡答題 (3分)

Describe three variations of the 'Delayed write' scheme used in distributed file systems.

你的答案

Write on ejection在驅逐時寫入。

Periodic write週期性寫入

Write on Close 關閉時寫入

### Q10 簡答題 (3分)

What are the three essential requirements (safety, liveness, and ordering) for Distributed Mutual Exclusion (DME) algorithms?

你的答案

分布式互斥演算法(DME)是為了確保多個獨立節點在共用臨界區CS時不會發生衝突。

### Q11 簡答題 (3分)

What is 'mobility transparency' in a DFS, and why is it important?

你的答案

當檔案或資料塊因為系統調度而在不同的實體儲存節點(伺服器)之間搬移時，使用者或應用程式完全不會察覺，且不需要修改存取路徑或停止運行

**Q12 簡答題** (3分)

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How does 'file replication' enhance a DFS in terms of scalability and fault tolerance?

你的答案

在FSD中檔案複製是指將同一個檔案的多個相同副本智儲存在不同的實體節點上。

- 1,提升可擴性：分散讀取流量,增強系統吞吐量。
- 2,提升容錯性：清除單點故障，確保資料高可用性。

**Q13 簡答題** (3分)

---

Compare and contrast 'one-copy update semantics' (UNIX) with 'session update semantics' (AFS) in DFS.

你的答案

單一副本更新語義UNIX：每一個寫入操作的結果必須立即可見。

會更更新語意(AFS)：只有在關閉更新過的檔案後，寫入操作的結果才會變得可用。

**Q14 簡答題** (2分)

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What is the difference between 'location transparency' and 'location independence' in DFS naming schemes?

你的答案

存取透明度：本地檔案與遠端檔案使用相同的API，程式不需要知道檔案是否分散在遠端。

位置透明度：在檔案重新定位時,用戶程式和用戶節點中的管理配置都保持不變。

**Q15 簡答題** (3分)

---

Describe two common 'cache location policies' and their respective benefits.

你的答案

伺服器記憶體：實作單，對Client透明。

客戶端記憶體，存取速度最快，可大幅提升效能。

客戶磁碟：容量較大，可快取更多更大的檔案，也能比較持久保存。

**Q16 簡答題** (2分)

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Compare 'write-through' and 'delayed write' schemes for cache update policies, highlighting their trade-offs.

你的答案

時即時寫入：優點：高一資料一致性,容錯性高。缺點：寫入效能低。

延遲寫入：優點：寫入效能極高。缺點：資料遺失風險。

**Q17 簡答題** (4分)

In HDFS, distinguish between the 'NameNode' and 'DataNodes' and their primary responsibilities.

你的答案

HDFS中的NameNode負責文件系統命名空間管理，名稱到文件的對應、處理開啟、關閉和重命名等操作。

DataNode：數據塊的分配/釋放

**Q18 簡答題** (2分)

What is the total message footprint per critical section invocation in the Ricart-Agrawala algorithm for a system of N processes?

你的答案

最壞情況：優先最低級的進程發選舉，需要 $O(N^2)$ 。的消息

最佳情況：最終的領導者發起選舉，需要 $O(N)$ 的消息。

**Q19 單選題** (2分)

The difference between the current readings of two physical clocks is referred to as:

A. Clock drift

B. Clock skew

你的選擇

C. Round-trip time

D. Latency

你的答案: B

**Q20 單選題** (2分)

A directory in a file system primarily maps:

A. IP addresses to MAC addresses

B. Text names to internal file identifiers

你的選擇

C. Memory addresses to disk sectors

D. Usernames to passwords

你的答案: B

**Q21 單選題** (2分)

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What is the primary advantage of file striping in a distributed file system like Lustre?

- A. It saves physical disk space.
- B. It enables parallel I/O accesses across servers.
- C. It automatically encrypts the data.
- D. It removes the need for metadata.

你的選擇

你的答案: **B**

**Q22 單選題** (2分)

---

If the replication factor in HDFS is reduced, what action does the NameNode take?

- A. It forces the client to crash.
- B. It commands DataNodes to clone blocks.
- C. It automatically selects and deletes the excess replicas.
- D. It formats the drive.

你的選擇

你的答案: **C**

**Q23 單選題** (2分)

---

Which of the following best characterizes a 'file' in a computer system?

- A. Highly volatile data in RAM.
- B. A named object immune to temporary system failures.
- C. A resource that cannot be shared.
- D. An entity without metadata.

你的選擇

你的答案: **B**

**Q24 單選題** (2分)

---

Which technology is primarily used to broadcast Coordinated Universal Time (UTC) globally?

- A. Undersea Ethernet cables
- B. Fiber optics
- C. Satellites (like GPS) and shortwave radio
- D. Bluetooth beacons

你的選擇

你的答案: **C**

**Q25** 單選題 (2分)

---

Which file system layer is closest to the application programs?

- A. Logical file system
- B. Basic file system
- C. I/O control
- D. Devices

你的選擇

你的答案: **A**

**Q26** 單選題 (2分)

---

Which system model relies on absolute global time to ensure that a read always returns the latest write?

- A. HDFS Append-only semantics
- B. DFS with Unix semantics
- C. AFS Session semantics
- D. Copy-on-write semantics

你的選擇

你的答案: **B**

**Q27** 單選題 (2分)

---

Distributed synchronization mechanisms are fundamentally designed to manage:

- A. Cooperative or competitive resource sharing
- B. CPU temperatures
- C. Hard drive formatting
- D. Monitor refresh rates

你的選擇

你的答案: **A**

**Q28** 單選題 (2分)

---

A file system provides APIs primarily to:

- A. Free programmers from storage allocation details
- B. Encrypt network traffic automatically
- C. Generate graphical UIs
- D. Compile source code

你的選擇

你的答案: **A**



**Q29** 單選題 (2分)

---

Which of the following is NOT a typical distributed synchronization mechanism?

- A. Clock synchronization
- B. Mutual exclusion
- C. Virtual memory paging
- D. Leader election

你的選擇

你的答案: C

**Q30** 單選題 (2分)

---

According to the HDFS architecture, where is the actual data of a file physically stored?

- A. NameNode
- B. DataNodes
- C. Client RAM
- D. Backup node

你的選擇

你的答案: B

**Q31** 單選題 (2分)

---

What is the primary role of a directory in a file system?

- A. To encrypt files
- B. To map text names to internal file identifiers
- C. To compress data blocks
- D. To format physical disks

你的選擇

你的答案: B

**Q32** 單選題 (2分)

---

What is the primary purpose of the replication factor in HDFS?

- A. To speed up the CPU clock
- B. To provide fault tolerance and data availability
- C. To reduce the physical disk space used
- D. To encrypt data over the network

你的選擇

你的答案: B

**Q33** 單選題 (2分)

---

The fact that crystals in different computers oscillate at slightly different frequencies causes:

- A. Clock skew
- B. Clock drift
- C. Deadlocks
- D. Race conditions

你的選擇

你的答案: **B**

**Q34** 單選題 (2分)

---

What does the metadata of a file typically NOT contain?

- A. The length of the file
- B. The owner's identity
- C. The actual user data
- D. Creation timestamps

你的選擇

你的答案: **C**

**Q35** 單選題 (2分)

---

Which type of transparency ensures that client programs are completely unaware of the distribution of files and use the same APIs for local and remote files?

- A. Location transparency
- B. Access transparency
- C. Scaling transparency
- D. Mobility transparency

你的選擇

你的答案: **B**

**Q36** 單選題 (2分)

---

What is a notable advantage of using client disk caching over client memory caching?

- A. It can easily support standard UNIX-like file-sharing semantics
- B. It is simpler to implement and completely transparent to clients
- C. It can cache much larger files and helps protect clients from server crashes
- D. It provides a higher performance speedup for transient local updates

你的選擇

你的答案: **C**

**Q37** 單選題 (2分)

---

Which variation of the delayed write scheme involves writing data back to the server specifically using session semantics?

- A. Write on ejection
- B. Periodic write
- C. Write-through caching
- D. Write on close

你的選擇

你的答案: **D**

**Q38** 單選題 (2分)

---

Under client-initiated cache validation policies, which checking variation aligns directly with standard Unix semantics?

- A. Checking before every access
- B. Checking when opening a file
- C. Checking only upon a client reboot
- D. Periodic checking at fixed intervals

你的選擇

你的答案: **A**

**Q39** 單選題 (2分)

---

Which protocol requires a write operation to be synchronously propagated to all replicas in the system, making it highly efficient when the read/write ratio is large?

- A. Primary-Copy Protocol
- B. Read One Write All Protocol (R1Wn)
- C. Available-Copies Protocol
- D. Quorum Consensus Protocol with weighted voting

你的選擇

你的答案: **B**

**Q40** 單選題 (2分)

---

In the Available-Copies Protocol, what must a server do immediately upon recovering from a site crash?

- A. Lock all available replicas across the network before answering any reads
- B. Promptly assign itself as the primary metadata node
- C. Bring itself up to date by copying from other servers before accepting user requests
- D. Flush its local client caches back to the primary server

你的選擇

你的答案: **C**

**Q41** 單選題 (2分)

---

During a write operation in the Quorum Consensus (QC) Protocol, how is the new version number determined?

- A. The server randomly generates a new version hash
- B. The client fetches all  $n$  copies and computes an average version number
- C. The client gets the copy with the largest version number from the  $w$  retrieved copies and increments it
- D. The primary node sets the version number to match the current system timestamp

你的選擇

你的答案: **C**

**Q42 單選題** (2分)

---

Which of the following is a characteristic of a stateless file server?

- A. It tracks clients' state information from one access request to the next
- B. Requests must be self-contained, identifying the file and exact position**
- C. It requires explicit open() and close() operations from clients
- D. It natively supports server-initiated cache validation

你的選擇

你的答案: **B**

**Q43 單選題** (2分)

---

Which of the following requires a stateful service architecture to function properly?

- A. Self-contained read/write requests
- B. File locking and server-initiated cache validation**
- C. Mounting remote directories onto local file trees via NFS v3
- D. Iterative pathname translations

你的選擇

你的答案: **B**

**Q44 單選題** (2分)

---

In a simple file service architecture, what is the core responsibility of the Flat File Service?

- A. Mapping logical text names of files to Unique File Identifiers (UFIDs)
- B. Intercepting user-level system calls in the client kernel
- C. Managing file directory files as structural client modules
- D. Handling operations and accesses directly on file contents using UFIDs**

你的選擇

你的答案: **D**

**Q45 單選題** (2分)

---

How are pathnames translated into file handles in the Network File System (NFS)?

- A. The client transmits the entire multi-part pathname to a single server for central resolution
- B. Pathnames are translated in an iterative manner step-by-step by the client module**
- C. The NameNode intercepts the pathname and pushes it to data slaves
- D. Pathnames are entirely bypassed by using flat 96-bit hard links

你的選擇

你的答案: **B**

**Q46** 單選題 (2分)

---

What role does the bio-daemon play in an NFS client implementation?

- A. It acts as a security manager checking user access control lists
- B. It executes remote procedure calls to break callbacks from stateful servers
- C. It handles pathname translations by looking up text identifiers
- D. It carries out asynchronous reads (read-ahead) and writes (delayed-write) to block input-output

你的選擇

你的答案: **D**

**Q47** 單選題 (2分)

---

What are the two distinct design characteristics that allow the Andrew File System (AFS) to achieve high scalability?

- A. Flat file operations and stateless server communication
- B. Whole-file serving and whole-file caching
- C. Append-only updates and multi-rack block striping
- D. Server-side pathname translation and primary-copy updates

你的選擇

你的答案: **B**

**Q48** 單選題 (2分)

---

AFS's design strategy relies heavily on which empirical assumption about UNIX file systems?

- A. Random file access is much more common than sequential access
- B. Files are typically very large, averaging over 64 megabytes
- C. Read operations on files are much more common than writes (approximately 6:1 ratio)
- D. Files are continuously shared and edited simultaneously by dozens of active users

你的選擇

你的答案: **C**

**Q49** 單選題 (2分)

---

In AFS, what is a 'callback promise'?

- A. A token issued by Vice guaranteeing to notify Venus if another client modifies a file
- B. A client-side message indicating data was successfully written to local disk
- C. A cryptographic ticket used to look up a 96-bit file identifier
- D. A guarantee that a file lock will never automatically expire

你的選擇

你的答案: **A**

**Q50** 單選題 (2分)

---

In the Hadoop Distributed File System (HDFS) master/slave architecture, what is managed exclusively by the NameNode?

- A. Serving file read and write operations directly to clients
- B. Storing and managing physical data block sequences
- C. File system namespace management and block allocation mapping
- D. Performing local block checksum validation routines

你的選擇

你的答案: **C**

**Q51** 單選題 (2分)

---

Which of the following describes the update/write rules for files in HDFS?

- A. Files support multi-user concurrent random modifications
- B. Files are write-once only, and all new writes are appended to the end of the file
- C. Files use a copy-on-write model synchronized via session semantics
- D. Files are broken into objects managed via Object Storage Targets

你的選擇

你的答案: **B**

**Q52** 單選題 (2分)

---

What is the purpose of the EditLog component in HDFS?

- A. It contains the entire namespace checkpoint image from the previous session
- B. It stores cached file blocks inside the client's local memory space
- C. It records every structural change made to file system metadata since the last FsImage checkpoint**
- D. It tracks bad block checksum alerts across individual DataNodes

你的選擇

你的答案: **C**